EART11300 Practical Skills Semester 2 (Chemistry 2)

Microscale gravimetric titration to determine water hardness

Health and Safety

Hazard	Risk	Control Measures
0.01M EDTA 	Causes serious eye irritation	Eye protection, lab coat and gloves always worn 
pH10 standard buffer solution	Not hazardous	
Eriochrome Black T indicator 0.2g solid  20 ml ethanol 	Eriochrome Black T causes serious eye irritation. Ethanol is highly flammable and causes serious eye irritation	Eye protection, lab coat and gloves always worn  Keep indicator away from heat/ignition sources
Glassware	Breakages lead to risk of cuts	Check apparatus for defects before use. Exercise care especially during clearing away/if glassware is wet. Follow correct disposal procedure for broken objects.

ILOs

By the end of this practical you should be able to:

- Carry out microscale gravimetric titration to determine the concentration of a chemical species (Ca^{2+} and Mg^{2+}) in solution
- Use molar ratios and quantitative molar relationships to calculate amounts of substance in terms of mass and concentration
- Compare your results with a UK national framework to assess and classify the hardness of water samples
- Evaluate the uncertainties in your measurements

The ILOs develop ideas from your EART11300 fieldwork at Mam Tor last semester considering how local geology affects water chemistry. This practical is also intended to develop safe working practices in the laboratory in preparation for the practical tasks in the second-year units EART26201 PRINCIPLES OF GEOCHEMISTRY and EART27001 ENVIRONMENTAL CHEMISTRY 1.

Background

Accurate quantitative analysis of the amounts (i.e. concentrations/masses) of species in solution can be carried out using a volumetric technique (titration). With a growing emphasis on sustainability, methods have been devised to carry out these procedures in microscale to conserve resources although these techniques may result in a larger error in our measurements. The

hardness of drinking water supply can be measured by titrating a water sample with 0.01M **e**thylene**d**iamine**t**etraacetic **a**cid (**EDTA**) solution at pH 10. Results can be assessed against national frameworks to identify potential impacts on local populations.

Aim: To measure and assess the total water hardness ($Ca^{2+}$and Mg^{2+}) in samples of water from different localities

Theory (the chemistry!):

What is water hardness?

Water hardness is caused by the presence Ca^{2+} and Mg^{2+} ions in water supply. As water from precipitation percolates through deposits that contain a high proportion of calcium or magnesium-rich minerals, divalent metal cations ($Ca^{2+}$and Mg^{2+}) may dissolve in the water. Local geology therefore affects water hardness. Criteria for assessing water hardness are not universal but frameworks often refer to the concentration of $CaCO_3$ or Ca^{2+} measured in milligrams per litre of water (ppm). United Utilities[1] and The Drinking Water Inspectorate[2] for the UK define degrees of water hardness as follows:

mg $CaCO_3$/l	**mg Ca^{2+}/l**	**Classification**
0-50	0-20	Soft
51-100	21-40	Moderately soft
101-150	41-60	Slightly hard
151-200	61-80	Moderately hard
201-300	81-120	Hard
>300	>120	Very Hard

When hard water is heated, thermal decomposition of bicarbonates can lead to deposition of solid calcium and magnesium compounds (scale) which may block pipes or affect the performance of water heating appliances. Conversely, soft water corrodes metal pipework at a faster rate than hard water[1]. Consuming hard water may have benefits for human health[3] but currently no guideline values are proposed for drinking water by the World Health Organization (WHO)[4]. Levels found in drinking water are not considered a health concern but water with [Ca^{2+}] > 100 mg/l can have a strong taste which might affect acceptability[4].

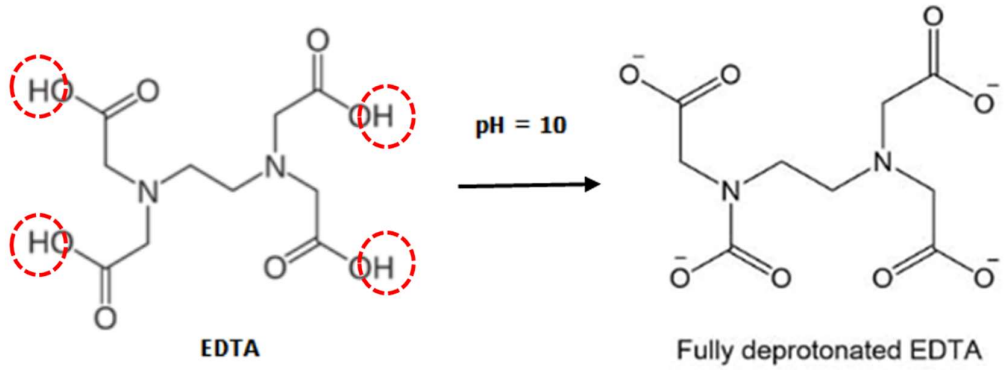
The gravimetric method

In this procedure water samples are tested using a dilute solution (0.01M) of EDTA. We can therefore assume that the solution and samples have the same density as pure water (1 gcm^{-3}). Using a fine-tip Pasteur pipette to deliver small volumes of our standard solution (EDTA), we can measure the total volume transferred during the titration by recording mass changes i.e. the mass of EDTA solution required in grams is numerically equivalent to the volume transferred in cm^3 since the density of a dilute solution is assumed to be the same as pure water.

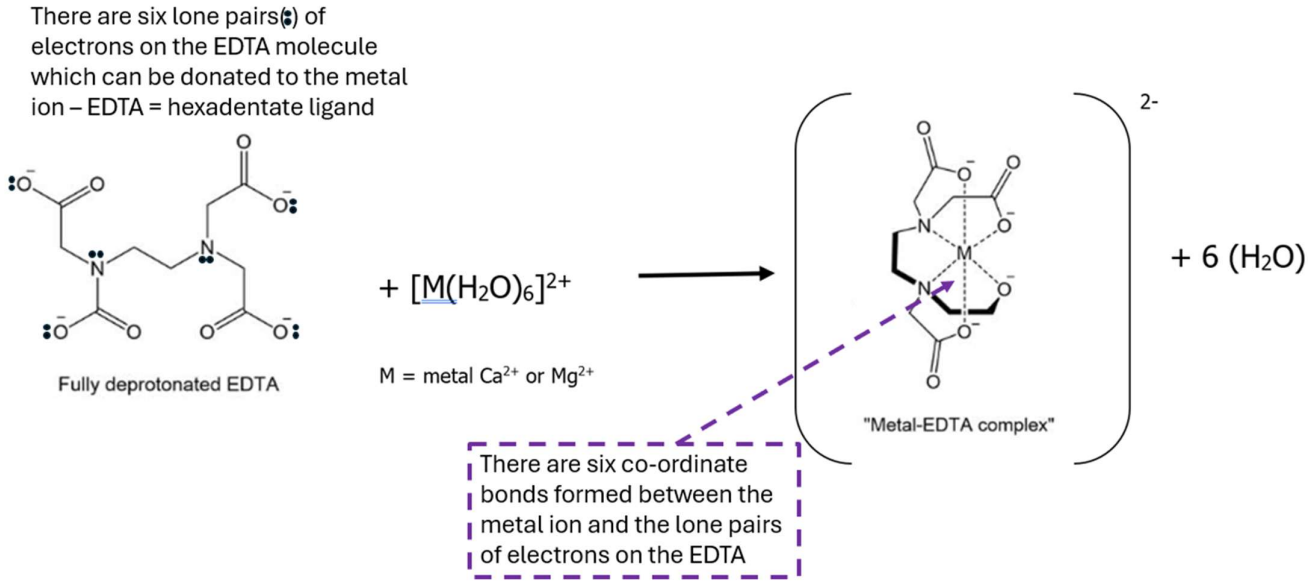
The titration complexation reaction

The structure of the EDTA molecule is shown below. At pH = 10 the EDTA molecule is in its fully deprotonated form (i.e. it has donated all its acidic hydrogen ions to the aqueous solution).

4 acidic hydrogens in carboxylic acid groups within the EDTA molecule.
The high pH means the EDTA exists in its deprotonated form



Fully deprotonated EDTA can bind to metal cations to form water-soluble complexes.



EDTA is a **hex**adentate ligand which means it has **six** lone pairs of electrons which can be donated to metal cations. The metal cation (**M** = Ca^{2+} or Mg^{2+} in this case) is enclosed within EDTA-complex and the process of complex formation releases smaller monodentate ligands (i.e. the 6 x water molecules) into the solution. This reaction is driven by the large increase in entropy as the number of particles in the system is increased when proceeding from the reactant side (2 particles) to the product side (7 particles). This phenomenon is known as the **chelate effect** (from Greek *chelos* = 'crab' because of the way in which the polydentate ligand figuratively wraps claws around the metal cation).

Note the reaction ratio between EDTA and the dissolved metal ion M (Ca^{2+} or Mg^{2+}**) is 1:1**.

The Indicator Reaction

Indicators are added to show when the end point of the reaction has been reached. Eriochrome Black T appears **red-purple** in the presence of Mg^{2+} but will turn **blue** once the EDTA in the standard solution has formed complex ions with all the calcium and magnesium ions in the water sample.

Procedure

Record all masses to 2 decimal places in the results table below.

1. Place a 10.5 ml glass sample vial onto the balance and record the mass of the empty vial in column 1 of the results table.
2. Select one of the waters from the stock samples to investigate and decant a small portion (between 2 – 5 ml) into a small beaker. With the sample vial still on the balance, use a dropping pipette to add ~2.5 g of the water to be tested into the vial and record the mass in column 2 then work out the mass of water added in column 6.
3. Use a dropping pipette to add ~0.2 g pH 10 standard buffer solution to the water sample.
4. Add two drops of Eriochrome Black T indicator from the dropping bottle provided to the water sample and swirl the vial carefully to mix the solutions.
5. Record the total mass of your water sample mixture in column 3 of the results table.
6. Remove the vial from the balance and place it over a piece of white paper – this enables the colour change at the end point to be identified easily.
7. Draw 0.01M EDTA solution into a fine-tip Pasteur pipette.
8. Add one drop of EDTA solution to the vial and swirl carefully to mix. Observe the colour of the indicator.
9. Continue adding drops of EDTA and swirling the mixture until the end point of the experiment identified when the indicator changes from **red-purple to blue**.
10. At the end point, record the new total mass of your mixture in column 4 and use this measurement to calculate the mass of EDTA added in column 5.
11. Repeat the experiment (recording results in a new row of the results table) to check for concordant amounts of EDTA added. Ideally, three concordant results should be obtained and an average calculated. If time is an issue, you could share results with your peers to confirm repeatability.
12. Repeat this process for additional water samples and record your results in the table.

Results Table

Water sample	**Recorded Mass / g (2dp)**				**Calculated masses /g (2dp)**	
	1	2	3	4	**5**	**6**
	empty sample vial	water sample + sample vial	indicator+ buffer + water sample + vial	Total at end point	**EDTA added** **(column 4 – column 3!)**	**Water sample** **(column 2 – column 1!)**
1	*8.68*	*11.19*	*11.46*	*12.32*	*0.86*	*2.51* *(11.19 - 8.68)*
1 (repeat)	*8.62*	*11.14*	*11.39*	*12.26*	*0.87*	*2.52*
2	*8.75*	*11.26*	*11.58*	*11.73*	*0.15*	*2.51*

Data Analysis

Useful molar quantity relationships:

Equation 1

$$Concentration\ (M) = \frac{moles\ (mol)}{Volume\ (l)}$$

Equation 2

$$Mass\ (g) = A_r(gmol^{-1}) \times moles\ (mol)$$

Remember to **check units**! You will need to convert cm^3 to litres (l or dm^3) and your final answers will be reported in milligrams per litre.

1. Use the concordant mass recordings from your table, and assuming the concentration of EDTA was sufficiently dilute that we can assume its density is the same as pure water (1 gcm^{-3}), calculate the mean volume of EDTA added to reach the end point for each of the water samples tested:
   e.g.
   Average mass (0.86 + 0.87 + 0.84) / 3 = 0.85666 = 0.86g

   Since the solution is dilute, we can assume its density is 1 gcm^{-3} (as pure water) and so the average volume of EDTA added is 0.86 cm3 (= 0.00086 litres)

2. Calculate the moles of 0.01M EDTA added to reach the end point for each of the samples tested.
   We know the volume of EDTA (from question 1 above = 0.00086 litres) and we know the concentration of EDTA (=0.01M) so we can use equation 1 rearranged to make moles the subject:

   Moles EDTA = concentration EDTA x volume EDTA
   moles = 0.01 x 0.00086 = 8.6 x 10^{-6} mol EDTA

3. Using the reacting ratio in the theory information above, determine the number of moles of Ca^{2+} + Mg^{2+} in the water sample.
   From the complexation equation the ratio of EDTA to metal ion is 1:1
   So the moles of Ca^{2+} + Mg^{2+} must be equal to the moles of EDTA i.e. = 8.6 x 10^{-6} mol Ca^{2+} + Mg^{2+}
4. Calculate the concentration of Ca^{2+} + Mg^{2+} in moles per litre:
   The mass of water (calculated in column 6 of the results tables) = 2.51g so the volume of water used can be assumed to be 2.51cm^3 (=0.00251 litres) since the sample is dilute. This amount of water contains 8.6 x 10^{-6} mol Ca^{2+} + Mg^{2+} (from part 3 above)

   Using equation 1 in terms of concentration
   Concentration (Ca^{2+} + Mg^{2+}) = moles Ca^{2+} + Mg^{2+} **/ volume water**
   = (8.6 x 10^{-6}) / (0.00251) = 3.426… x 10^{-3} mol/l

5. Convert the concentration in moles per litre to milligrams per litre (**NOTE**: an important assumption is made here; we assume all the hardness is due to calcium ions (**!**) so the atomic mass of calcium is commonly used and not some average of the atomic masses of Ca and Mg as we do not know their relative proportions in the sample. A_r(Ca) = 40.078u

   Equation 2 shows mass and moles are related by Ar. This means we can convert molar concentrations (as found in step 4 above) to mass concentrations simply by multiplying the molar concentration by the A_r (in this case of Ca)
   Mass concentration = 3.426… x 10^{-3} x 40.078 = 0.13731885 g/l
   But the answer will be in grams so we now need to convert to milligrams (x1000)
   Mass concentration = **137 mg/l (compare with sample mineral content lable)**

6. Using the framework of United Utilities/Drinking Water Inspectorate in the background section above, assess and classify the hardness of your water samples.

   137 mg/l **>** 120 mg/l therefore this sample would be classified as "very hard" within this framework

7. Can you make any deductions about the geology of the water sample source regions?
   Possibly the local geology at source consists of calcium/magnesium-rich minerals

Uncertainties in the measurement using the microscale technique

1. The mass balance measures mass to 2 decimal places, so the absolute uncertainty of the instrument is ±0.005g. A difference of two masses is used to find the mass of EDTA solution transferred so this uncertainty must be considered twice (i.e. the uncertainty is 2 x ±0.005g = ±0.01g).
   The fine-tip Pasteur pipette can deliver a drop of solution with a volume of 0.02 ml (i.e. 0.02cm^3). Assuming the solution is very dilute, this corresponds to a mass of 0.02g of solution. It may be that one drop contains more moles of EDTA than required to complex with the remaining metal ions and so the mass of EDTA added is more than required. This could be up to almost the entire mass of the drop so we can say that the uncertainty introduced by the volume of the drop is up to a maximum of +0.02g.
   The uncertainty in our measurement is asymmetric: -0.01g and +0.03g
   Using one result for each of the water samples tested, calculate the percentage errors in each of the measurements for the upper and lower bounds.

   e.g. first result for sample 1 from results table above EDTA = 0.86g

   lower bound % error = (-0.01g/0.86) x 100 = 1.2 %

   Assuming the end point was identified with the addition of one drop of EDTA (+0.02g) and difference of two masses introduces uncertainty of ±0.01g

   Upper bound % error = +(0.03/0.86) x 100 = +3.5%

2. Using the percentage errors determined in part 1. above, determine maximum and minimum values for the concentration of (Ca^{2+} + Mg^{2+}) in mg/l. Comment on these ranges – does the range affect how the water sample hardness could be classified?

-1.2 % of 137 = 135.4 mg/l
+3.5% of 137 = 141.8 mg/l
The range is 6.4 mg/l. The values for the lower/upper bounds would not change the classification of this water sample as they are all >120mg/l (the threshold for "very hard") However, the % error will increase for a water sample with a lower hardness (as less EDTA required for titration) and the range might be large enough that the values could span across two water hardness classes or the sample may be close enough to a class division that the error bars would span across the division.

3. In calculating the total water hardness, the atomic mass (A_r) of Ca was used instead of some weighted average of Ca and Mg. What is the impact of this assumption on the accuracy of your results – will your results be greater or less than the 'true' value? Explain your answer.
   We used the molar relationship in equation 2 (mass = A_r x moles) to convert from molar to mass concentration
   This means the mass concentration will be greater than the true value since factoring in the atomic mass for magnesium (which is considerably less than that of calcium - A_r(Mg) = 24.305) should reduce the atomic mass number used to convert between the molar and mass concentration.

4. Tap water may be delivered though old plumbing systems which incorporate lead piping. Some divalent lead ions (Pb^{2+}) may dissolve in the water supply. Pb^{2+} ions will combine with the Eriochrome Black T indicator molecule. If lead ions were present in a sample of tap water tested by this method, what effect would these have on your results for total water hardness? Explain your answer.
   The calculated total water hardness (mass concentration of Ca^{2+} + Mg^{2+}) would be greater than the true value because a larger volume of EDTA would be required to form complexes with the Ca2+, Mg2+ and Pb2+ ions than if no contamination from Pb2+ was present. This means that the moles of Ca2+ + Mg 2+ is greater than the true value and this would propagate through the calculation of mass concentration.

References and Further Information

[1] United Utilities, Water for the North West: Water hardness https://www.unitedutilities.com/help-and-support/your-water-supply/your-water/about-your-water/water-hardness/ (there is a PDF factsheet to download)

[2] The Drinking Water Inspectorate www.dwi.gov.uk

Water hardness https://dwi.gov.uk/consumers/learn-more-about-your-water/water-hardness-hard-water/ (includes showing the water hardness of drinking water supplies in England and Wales)

[3] *Calcium and magnesium in drinking-water: Public health significance*

Edited by J. Cotruvo and J. Bartram Published in 2009 by the World Health Organization A review of the contribution of drinking-water to total daily intake of calcium and magnesium, and an assessment of possible health benefits, including reducing cardiovascular disease mortality and osteoporosis.

https://www.who.int/publications/i/item/9789241563550

[4] *Guidelines for drinking-water quality: fourth edition incorporating the first and second addenda*.

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https://www.who.int/publications/i/item/9789240045064